API 650 TANK DESIGN

DESIGN, FABRICATION, INSPECTION, ERECTION & TESTING OF ABOVE GROUND WELDED STORAGE TANKS



WHAT IS API 650?

API 650 is an international standard used for design, fabrication, inspection, erection, and testing of Above Ground Welded Storage Tanks.

– MAINLY USED IN:

- Oil & Gas
- Chemical Plants
- Refineries
- Petrochemical Industries
- Water Storage Systems

PURPOSE OF API 650 TANK

Designed for:
- ✓ Atmospheric pressure storage
- ✓ Low pressure liquid storage

Typical Stored Liquids:
- Crude Oil
- Diesel
- Petrol
- Chemicals
- Water
- Solvents

BASIC DESIGN CONDITIONS

- Pressure : Atmospheric
- Temperature : Up to 93°C typically
- Tank Type : Vertical cylindrical
- Material : Carbon steel / Stainless steel

MAIN PARTS OF API 650 TANK

1. BOTTOM PLATE
Purpose:
- Supports liquid load
- Transfers load to foundation

Types:
- Lap welded bottom
- Annular plate bottom

2. SHELL
Main cylindrical body of tank.

Design based on:
- Hydrostatic pressure
- Wind load
- Seismic load
- Corrosion allowance

Shell thickness increases from top to bottom because pressure increases.

3. ROOF
Protects stored liquid.

Roof Types:
- Cone roof → Small tanks
- Dome roof → Large tanks
- Floating roof → Volatile liquids
- Internal floating roof → Vapor loss control

FLOATING ROOF PURPOSE

Used to reduce:
- ✓ Vapor loss
- ✓ Fire risk
- ✓ Emissions

Common in:
- Petrol
- Naphtha
- Crude oil tanks

IMPORTANT DESIGN PARAMETERS

1. **Tank Diameter** – Determined from storage capacity and plot space.
2. **Tank Height** – Depends on required volume, safety margin and pump suction requirement.
3. **Design Liquid Level** – Maximum liquid height considered for design.
4. **Specific Gravity** – Higher SG = Higher shell thickness required.
5. **Corrosion Allowance** – Extra thickness added for corrosion protection. Typical: 1.5 mm to 3 mm

API 650 DESIGN LOADS

- **Dead Load** – Tank self weight
- **Hydrostatic Load** – Liquid pressure load
- **Wind Load** – Very critical for tall tanks
- **Seismic Load** – Earthquake consideration
- **Roof Load** – Live load on roof
- **Nozzle Load** – Pipe connection forces

SHELL DESIGN CONCEPT

Hydrostatic Pressure Formula

$$P = \rho g h$$

Where:
- P = Pressure
- ρ = Density
- g = Gravity
- h = Liquid height

As height increases:
- ✓ Pressure increases
- ✓ Shell thickness increases

BOTTOM DESIGN

Bottom Plate Thickness:
- Usually 6 mm minimum
- Based on corrosion + load

ANNULAR PLATE

Used for:
- Large diameter tanks
- High shell stress areas

Provides:
- Better load distribution

ROOF DESIGN

Cone Roof:
- Simple
- Economical
- Easy fabrication

FLOATING ROOF DESIGN

Important components:
- Pontoon
- Deck
- Seal system
- Roof drain
- Rolling ladder

NOZZLE DESIGN

Tank nozzles include:
- Inlet
- Outlet
- Drain
- Vent
- Manhole
- Overflow

Design checks:
- Reinforcement pad
- Stress analysis
- Weld quality

FOUNDATION DESIGN

Tank foundation is extremely critical.

Foundation Types:
- Ring wall → Large tanks
- Earth foundation → Small tanks
- Pile foundation → Weak soil

TANK SETTLEMENT

Uneven settlement can cause:
- Shell cracking
- Bottom leakage
- Roof distortion

Monitoring required.

API 650 MATERIAL

Common materials:
- ASTM A36
- ASTM A283
- ASTM A516

Selection based on:
- Temperature
- Corrosion
- Product compatibility

WELDING REQUIREMENTS

Important welds:
- Shell vertical weld
- Shell circumferential weld
- Bottom weld
- Roof weld

INSPECTION & TESTING (NDT METHODS)

- RT (Radiography)
- UT (Ultrasonic)
- MPI (Magnetic Particle Inspection)
- Vacuum Box Test

HYDROTEST

Tank filled with water for testing.

Purpose:
- Leak test
- Settlement observation
- Structural integrity verification

CORROSION PROTECTION

Methods:
- Painting
- Internal coating
- Cathodic protection
- Linings

IMPORTANT API 650 CALCULATIONS

Tank Volume Formula:

$$V = \pi r^2 h$$

Shell Stress Formula:

$$\sigma = \frac{PD}{2t}$$

Where: P = Pressure, D = Diameter, t = Thickness

API 650 VS API 620

API 650		API 620
Atmospheric tank	VS	Low pressure tank
Lower pressure		Higher pressure
Simpler design		More complex
Common in refinery		Special applications

PRACTICAL MAINTENANCE/CONCERNS | **INSPECTION STANDARDS RELATED** | **REAL INDUSTRY FOCUS**

ASME CODES

ASME B31 – PIPING CODES



B31.1	Power Piping – For power plants & general services
B31.2	Fuel Gas Piping – For fuel gas systems
B31.3	Process Piping – For chemical, petrochemical & refinery plants
B31.4	Pipeline Transportation Systems – For liquid & gas pipeline systems
B31.8	Gas Transmission & Distribution Piping Systems – For gas transmission & distribution
B31.9	Building Services Piping – For HVAC, plumbing & building systems
B31.12	Hydrogen Piping & Pipelines – For hydrogen service systems

ASME B16 – COMPONENT CODES



B16.1	Cast Iron Pipe Flanges & Flanged Fittings
B16.5	Pipe Flanges & Flanged Fittings
B16.9	Factory-Made Wrought Buttwelding Fittings
B16.11	Forged Fittings, Socket-Welding & Threaded
B16.20	Metallic Gaskets for Pipe Flanges
B16.25	Buttwelding Ends
B16.34	Valves – Flanged, Threaded & Welding End
B16.47	Large Diameter Steel Flanges (NPS 26 & larger)
B16.48	Line Blanks
B16.49	Fabricated Steel Pipe Flanges

ASME B36 – PIPES & TUBES



B36.10M	Welded & Seamless Wrought Steel Pipe
B36.19M	Stainless Steel Pipe
B36.22M	Copper & Copper Alloy Seamless Tube
B36.23M	Copper & Copper Alloy Welded Tube
B36.24M	Stainless Steel Seamless Tube
B36.25M	Nickel Alloy Seamless Pipe & Tube

BOILER & PRESSURE VESSEL CODES

Section I	Power Boilers
Section II	Material Specifications
Section IV	Heating Boilers
Section VIII	Pressure Vessels
Division 1	Mandatory Appendix
Division 2	Alternative Rules



OTHER IMPORTANT ASME CODES

BPE	Bioprocessing Equipment
BPVC	Boiler & Pressure Vessel Code
B73.1	Flexible Metal Hose
BTH-1	Thermowells
PTC	Performance Test Codes
SFA	Welding Filler Metals
Y14.5	Dimensioning & Tolerancing
Y1.1	Engineering Drawings



MATERIAL & DESIGN CODES

Section II	Material Specifications
Section III	Rules for Construction of Nuclear Facility Components
Section IV	NDE (Non Destructive Examination)
Section VI	Recommended Rules for Care & Operation
Section VII	Recommended Guidelines for Plant Construction & Pressure Vessels

WHY ASME CODES ARE IMPORTANT



- ✓ Ensure Safety of People, Environment and Assets
- ✓ Improve Reliability & Operational Efficiency
- ✓ Ensure Regulatory Compliance Globally
- ✓ Reduce Risk, Downtime & Maintenance Costs
- ✓ Enable Interchangeability & Standardization
- ✓ Build Confidence. Deliver Excellence.

ASME PIPING CODES

ASME B3 series codes provide requirements for the design, materials, fabrication, and inspection of piping systems across various industries.

ASME CODE	DESCRIPTION	ASME CODE	DESCRIPTION
ASME B31.1	Power Piping	ASME B16.5	Pipe Flanges and Flanged Fittings
ASME B31.2	Fuel Gas Piping	ASME B16.9	Factory Made Wrought Butt-Welding Fittings
ASME B31.3	Process Piping	ASME B16.11	Forged Fittings, Socket-Welding and Threaded
ASME B31.4	Pipeline Transportation Systems for Liquid Hydrocarbon & other Liquids	ASME B16.25	Buttwelding Ends
ASME B31.5	Refrigeration Piping	ASME B16.34	Valves – Flanged, Threaded and Welding End
ASME B31.8	Gas Transmission & Distribution Piping System	ASME B16.47	Large Diameter Steel Flanges (NPS 26 and Larger)
ASME B31.9	Building Services Piping	ASME B16.48	Line Blanks
ASME B31.11	Slurry Transportation Piping Systems	ASME B16.49	Manufactured Fittings
		ASME B16.51	General Purpose Valves
		ASME B16.10	Face-to-Face & End-to-End Dimensions of Valves
		ASME B36.10	Welded and Seamless Wrought Steel Pipe
		ASME B36.19	Stainless Steel Pipe
		ASME B36.25	Buttwelding Long Radius Elbows and Returns

ASME SECTION IX

Welding, Brazing and Fusing Qualifications



1 Introduction

- ASME Section IX is the qualification standard for welding, brazing and fusing procedures and personnel.
- It ensures that welds are made by qualified procedures and qualified welders, to meet code requirements.
- Applicable to construction of pressure vessels (Section VIII), piping (Section I), nuclear components (Section III), and other code constructions.
- Works together with construction codes like ASME Section I, II, III, IV, V and ASME B31.1, B31.3 etc.

2 Key Terms

- ✓ WPS – Welding Procedure Specification
- ✓ PQR – Procedure Qualification Record
- ✓ WPQR – Welder Performance Qualification Record
- ✓ WQT – Welder Qualification Test
- ✓ WPQR – Welding Operator Performance Record
- ✓ WPS/PQR – Documents qualification procedure
- ✓ WPS is prepared from PQR.
- ✓ WPQR/WQT confirms welder/operator can produce acceptable welds as per qualified WPS.

3 Scope (What it Covers)

- ✓ Welding (all processes)
- ✓ Brazing
- ✓ Fusing (e.g. plastic piping)
- ✓ Qualification of procedures
- ✓ Qualification of welders, brazers and operators
- ✓ Essential and non-essential variables
- ✓ Ranges of qualification
- ✓ Use of Supplementary Essential Variables (SEVs)
- ✓ Requirements for change in WPS (QW-200)

4 Main QW Articles

Article	Title
QW-100	General Requirements
QW-200	Welding Procedure Specifications (WPS)
QW-300	Welder Performance Qualification (WPQ)
QW-400	Welding Variables
QW-500	Standard Welding Procedure Specifications (SWPS)
QW-600	Brazing
QW-700	Fusing
QW-800	Supplementary Essential Variables
QW-900	Qualification of Welding Operators

5 Qualification Documents

- **PQR (QW-200)** → Test weld is made and tested as per Section IX.
- **WPS (QW-200)** → WPS is prepared using qualified ranges from PQR.
- **WPQR / WQT (QW-300)** → Welder/Operator is tested using the qualified WPS.
- **Production Welding** → Welds are made in accordance with qualified WPS by qualified personnel.

6 Welding Positions (QW-461)



1G Flat	2G Horizontal	3G Vertical	4G Overhead
5G Pipe Horizontal (Fixed)	6G Pipe Inclined (45°)	6GR Pipe with Restrictor Ring	6G Pipe Vertical (Fixed)

7 Common Welding Processes

Process	Abbreviation
Shielded Metal Arc Welding	SMAW
Gas Metal Arc Welding	GMAW (MIG/MAG)
Gas Tungsten Arc Welding	GTAW (TIG)
Flux Cored Arc Welding	FCAW
Submerged Arc Welding	SAW
Electroslag Welding	ESW
Oxy-Fuel Gas Welding	OFW
Resistance Welding	RW
Stud Welding	SW
Brazing	BRAZING
Fusing (Plastic)	FUSING

8 Essential vs Non-Essential Variables (QW-200)

Essential Variables (A change requires requalification)
- Process
- P-Number (Group)
- Filler Metal (F-Number)
- Base Metal Thickness
- Welding Position
- Weld Type (Groove/Fillet)
- Backing
- Preheat / Interpass Temperature
- PWHT
- Gas / Flux
- Electrical Characteristics
- Technique (e.g. stringer/weave)
- Joint Design (within limits)

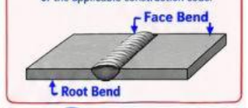
Non-Essential Variables (within qualified limits)
- Travel Speed
- Current / Voltage
- Electrode / Filler Size
- Joint Design (within limits)
- Coupling
- Number of Passes (as per limits)
- Cleaning Methods
- Details of Backing (within limits)
- Interpass Temperature (within limits)
- Other variables listed in QW-202

9 Qualification Ranges (Examples)

Variable	Qualification Range
Base Metal Thickness	From qualified thickness to unlimited (as per QW-451)
P-Number Group	Same P-Number or higher (as per QW-422)
Filler Metal (F-Number)	Same F-Number or higher (as per QW-404)
Welding Position	Positions qualified and all lower positions
Preheat / Interpass	Same or higher
PWHT	Same or smaller diameter (as per QW-452)
Pipe Diameter (for pipe)	Same or smaller diameter (within limits)
GTAW with/without filler	With filler qualifies without; but not vice versa (QW-408)

10 PQR Test Requirements (QW-200)

- ✓ Test weld as per proposed WPS.
- ✓ Tensile test (required).
- ✓ Guided bend test (required).
- ✓ Face bend and root bend (as applicable).
- ✓ Impact test (if required).
- ✓ Hardness test (if required).
- ✓ Macro / Micro examination (if required).
- ✓ All tests must meet acceptance criteria of the applicable construction code.



11 Acceptance Criteria

- ✓ Tensile: Meets minimum required strength.
- ✓ Bend: No open defects (as per code).
- ✓ Impact: Meets required energy (if applicable).
- ✓ Hardness: Within specified limits (if applicable).
- ✓ Macro: Acceptable fusion and penetration.
- ✓ Visual: Free from unacceptable defects.
- ✓ As per relevant construction code (e.g. Sec VIII, B31.3).

12 Change in WPS (QW-200)

- ✓ Change in essential variables requires requalification.
- ✓ Change in non-essential variables allowed within qualified limits.
- ✓ Use of Supplementary Essential Variables (SEVs) where permitted.
- ✓ Follow QW-200 and QW-202 for details.

13 Inspector / Engineer Key Points

- ✓ Verify WPS and PQR are qualified and approved.
- ✓ Check welder/operator qualification validity.
- ✓ Ensure welding within qualified ranges.
- ✓ Verify consumables (F-Number) and base metals (P-Number).
- ✓ Check preheat, interpass and PWHT.
- ✓ Review variable changes and SEVs.
- ✓ Ensure records are maintained.
- ✓ Witness and review mechanical test results.

14 Key Takeaways

- ✓ Follow the code, not assumptions.
- ✓ Qualification ensures quality, safety and reliability.
- ✓ Understand essential variables.
- ✓ Maintain proper documentation.
- ✓ Train and qualify your people.
- ✓ Quality welds build safe and sustainable assets.

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